Exp 4 · Code converters: binary → Gray, BCD → Excess-3

Manual pp. 11–12 · ICs 7486, 7432, 7408, 7404 · both converters checked over every valid input

2 Adders/subtractors → 3 NAND/NOR design → **4 Code converters** → 5 Comparator & parity

MANUAL EXP 4, AIM

Design and implement (A) a 4-bit binary to Gray code converter, (B) a BCD to Excess-3 converter. Viva form: "Why is Gray code used in position encoders, and why is Excess-3 self-complementing?"

GUESS FIRST

How many bits change between consecutive Gray codes? What is 9's complement of 2 in Excess-3?

A · Binary → Gray

$$G_3 = B_3, \qquad G_2 = B_3 \oplus B_2, \qquad G_1 = B_2 \oplus B_1, \qquad G_0 = B_1 \oplus B_0$$

Wiring (manual's pins):

- B3 → output G3 directly.
- 7486 gate 1: B3 → 1, B2 → 2 → **3 = G2**.
- 7486 gate 2: B2 → 4, B1 → 5 → **6 = G1**.
- 7486 gate 3: B1 → 9, B0 → 10 → **8 = G0**.

B	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
G	0000	0001	0011	0010	0110	0111	0101	0100	1100	1101	1111	1110	1010	1011	1001	1000

Gray → binary (exercise): B3 = G3, B2 = B3 ⊕ G2, B1 = B2 ⊕ G1, B0 = B1 ⊕ G0. It is a chain: each XOR uses the *previous output*, not the previous input.

B · BCD → Excess-3 (XS-3 = BCD + 3)

BCD ABCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001
XS-3 WXYZ	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100

The K-maps use 1010–1111 as don't-cares (they are not BCD). The manual's expressions, and a factored form that saves gates:

$$W = A + BC + BD = A + B(C + D) \qquad X = B'C + B'D + BC'D' = B \oplus (C + D)$$

$$Y = CD + C'D' = (C \oplus D)' \qquad Z = D'$$

Wiring (factored form):

Gate	Wires
7432 gate 1	C → 1, D → 2 → 3 (C + D)
7408 gate 1	B → 1, 7432-3 → 2 → 3 (B(C + D))
7432 gate 2	A → 4, 7408-3 → 5 → **6 = W**
7486 gate 1	B → 1, 7432-3 → 2 → **3 = X**
7486 gate 2	C → 4, D → 5 → 6; then 7404: 7486-6 → 1 → **2 = Y**
7404	D → 3 → **4 = Z**

TRAP: INPUTS ABOVE 1001

Set 1010–1111 and the outputs look like nonsense. Those inputs aren't BCD, and the design treated them as don't-cares. Test only 0000–1001.

WRITE IN YOUR RECORD

Result: the binary-to-Gray converter (G3 = B3, Gi = Bi+1 ⊕ Bi) and the BCD-to-Excess-3 converter (W = A + BC + BD, X = B'C + B'D + BC'D', Y = CD + C'D', Z = D') were implemented and their truth tables verified.

Theory for the record: a code converter changes one binary code into another. In Gray code, successive values differ in one bit only. That avoids false intermediate readings in shaft encoders, and it gives the order of K-map labels. Binary → Gray: $G_n = B_n$, $G_i = B_{i+1} \oplus B_i$. Excess-3 is BCD + 3. It is self-complementing: inverting all bits gives the 9's complement, which simplifies decimal subtraction. Each output bit is found from a K-map, with the six unused BCD codes as don't-cares.

CHECK 1

(a) Convert Gray 1011 to binary. (b) Show XS-3 is self-complementing using 2 and 7. (c) Why can X = B'C + B'D + BC'D' be written B ⊕ (C + D)?

Answers

Opening question. One bit changes between consecutive Gray codes, so a sensor caught mid-transition can be off by at most one step. 2 in XS-3 is 0101; inverting gives 1010 = XS-3 of 7 = 9 − 2. So the 9's complement is just the bitwise inverse.

Check 1. (a) B3 = 1, B2 = 1 ⊕ 0 = 1, B1 = 1 ⊕ 1 = 0, B0 = 0 ⊕ 1 = 1, giving **1101**. (b) XS-3(2) = 0101 and XS-3(7) = 1010: bitwise inverses, and 2 + 7 = 9. (c) B'C + B'D = B'(C + D), and BC'D' = B(C + D)'. B'P + BP' with P = C + D is exactly B ⊕ P.